3.4 *Processing*

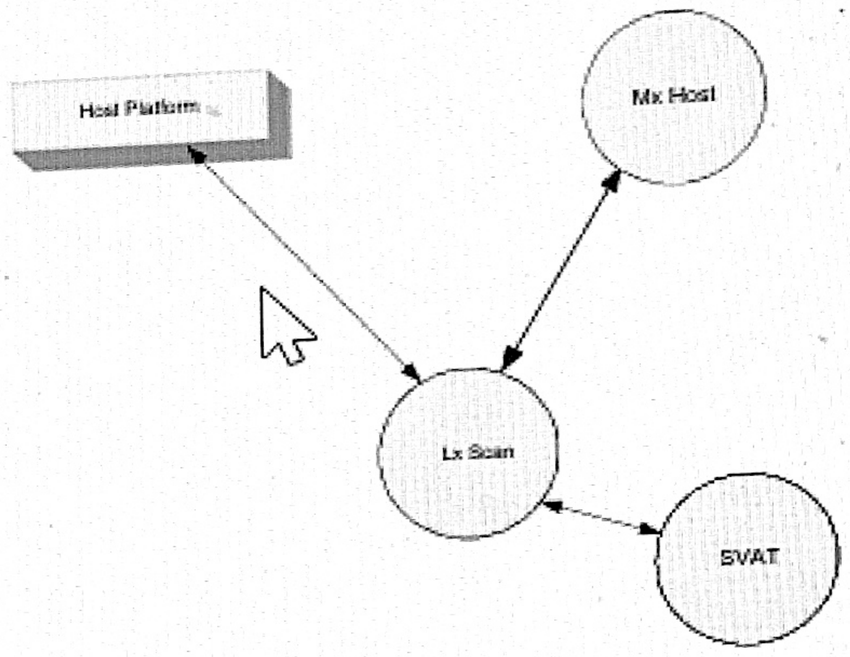
3.4.1 Key Processing

The following subsections define the key processing flows within the MR system related to the Host Software Subsystem. Each processing flow corresponds to a major step within the user operation of the system.

These processing flows are also used within a service session for system diagnosis and calibration. In this case the patient is the service ID. The operation can be either manual, under the service engineer's direct control, or automated, under the control of a service tool initiated by the service engineer.

3.4.1.1 Exam Initiation

Figure 12 Data Paths for Exam Initiation



Exam initiation is controlled within the host software subsystem. The following key actions occur to complete this processing step:

- Host platform allows operator to identify patient and protocols for study.
- Lx Scan performs following checks to ensure system ready for new exam
  - Required option keys are installed
  - Real time sub-systems is up
  - System is calibrated
  - No exam/diagnostic session is in progress
- Host platform sends selected patient information to Lx Scan

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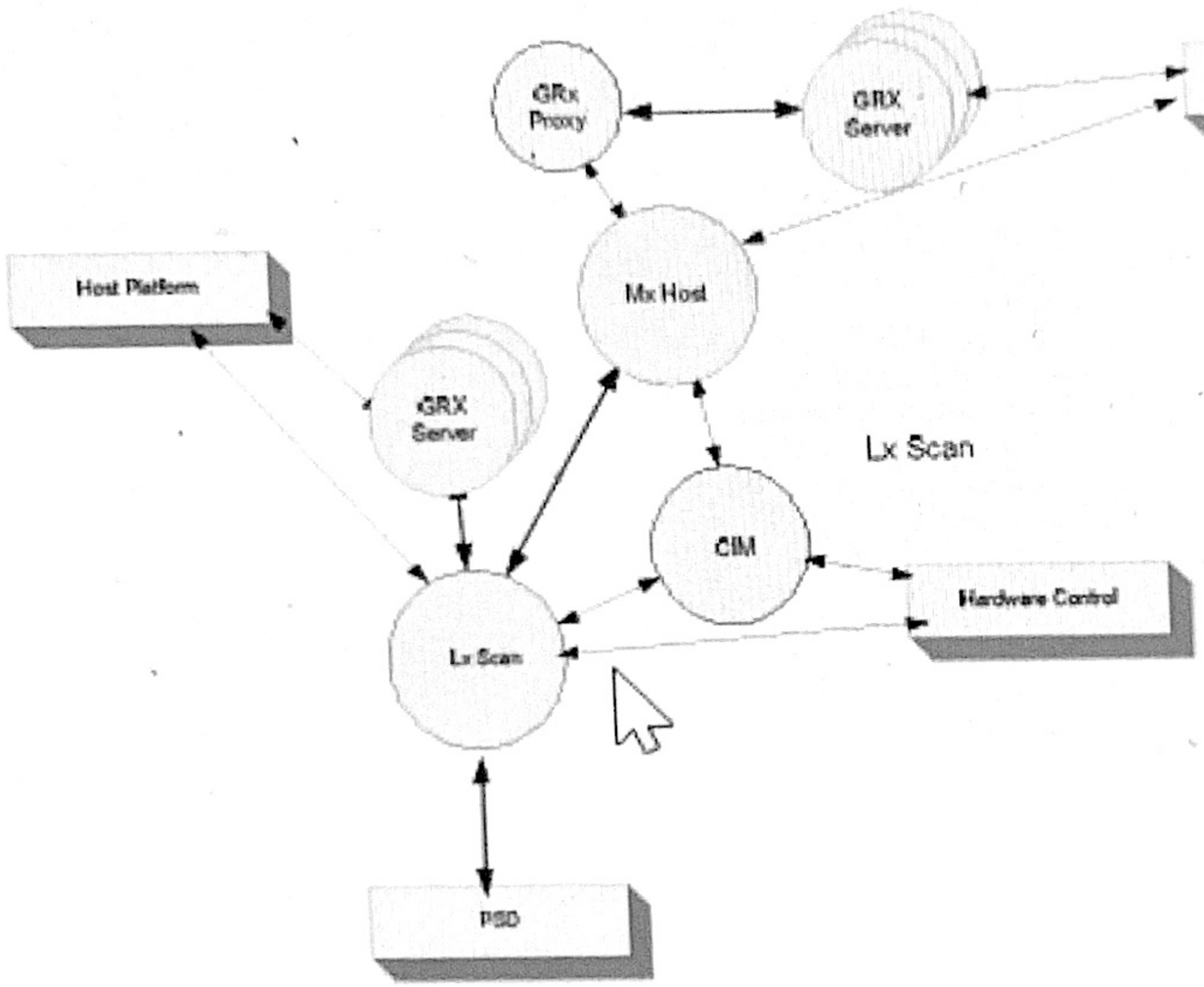
- After successful exam start, Lx Scan creates new exam object & fills in complete patient information
- Lx Scan sets the safety limits as requested by the user

3.4.1.2 Scan Prescription

Figure 9 Data Paths for Scan Prescription



Scan Prescription is controlled within the host software subsystem with significant interaction with the Pulse Sequence subsystem. The following key actions occur to complete this processing step:

- Host platform moves the task to InRx state and sends protocol path to Lx Scan.
- For Lx tasks, Lx Scan starts appropriate Lx Host PSD based on protocol
- For Mx tasks, Lx Scan requests Mx JVM to launch Mx application based on user selection or protocol
- For Lx tasks, Lx Scan requests coil information from CIM. For Mx tasks, Mx JVM requests coil information from CIM
- For dynamic coil configuration, CIM runs auto coil algorithm if full coil mode calibration available.
- System configuration and operator selections are sent to Host PSD
- For Lx tasks, Lx Scan creates the user interface for prescription based on inputs from Host PSD and Hardware control

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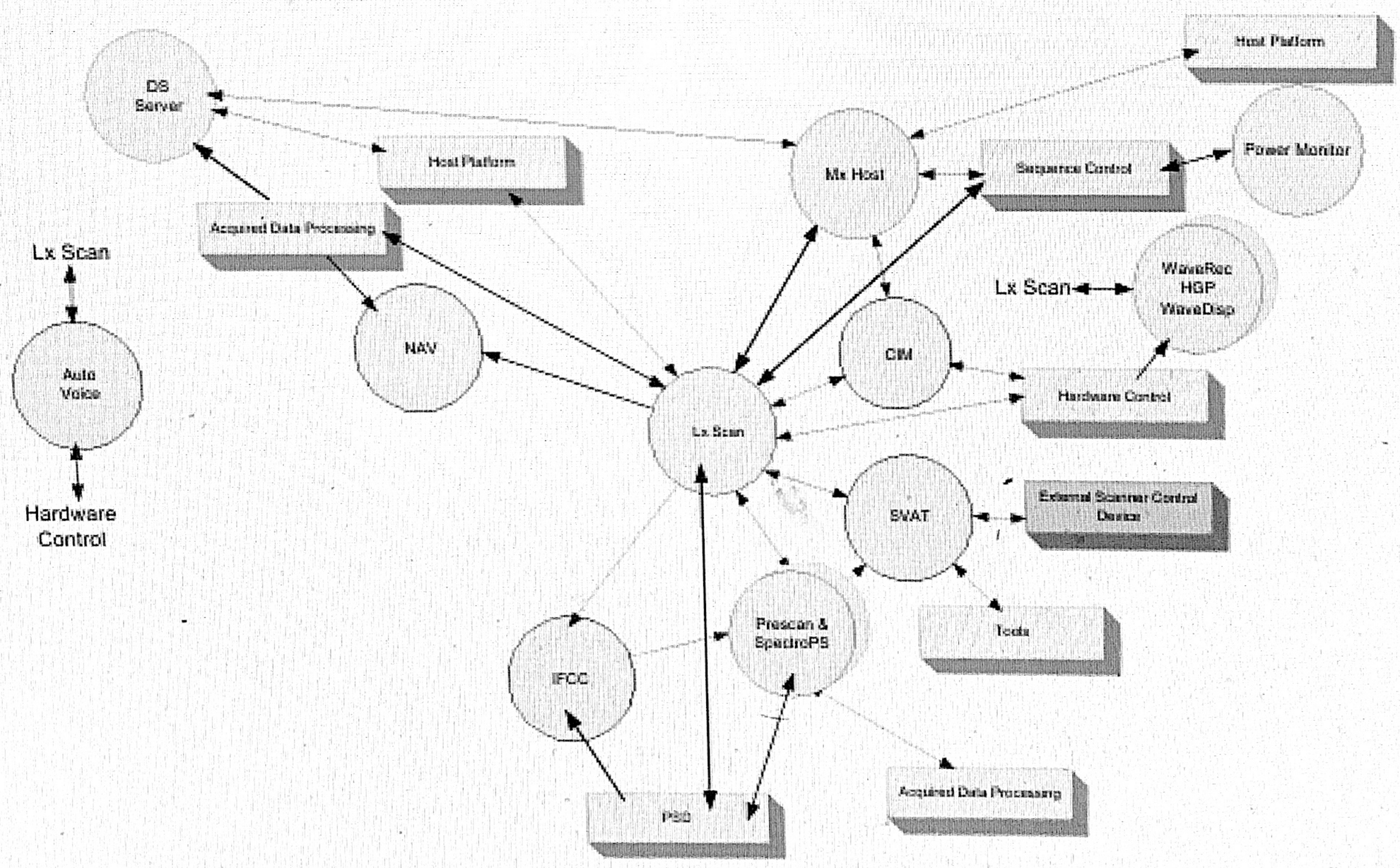
- For Mx tasks, Mx JVM shall create user interface for prescription based on inputs from Host PSD and Host Recon components.
- GRX servers provides graphical user interface in three viewports for prescribing scan locations
- User selections are validated with PSD and operator is guided with advisory values
    - Choose coils (Rx / Tx) as needed.
    - Choose gradient mode as needed.
    - Define scanning range (slices, slice groups, volumes...)
    - Define saturation bands as needed
    - Set imaging parameters
    - Define automated processing (subtractions, DTI etc.)

Once the prescription is complete the operator saves the prescription placing the task in the RXD state.

3.4.1.3 Scan Initiation (Download/Prep/Start)

Figure 10 Data Paths for Scan Initiation



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Scan initiation (download/prep/start) is initiated by the host software subsystem with the significant processing done within the PSD, Recon, Hardware Control and Sequence Control subsystems. The following key actions occur to complete this processing step:

For Lx tasks

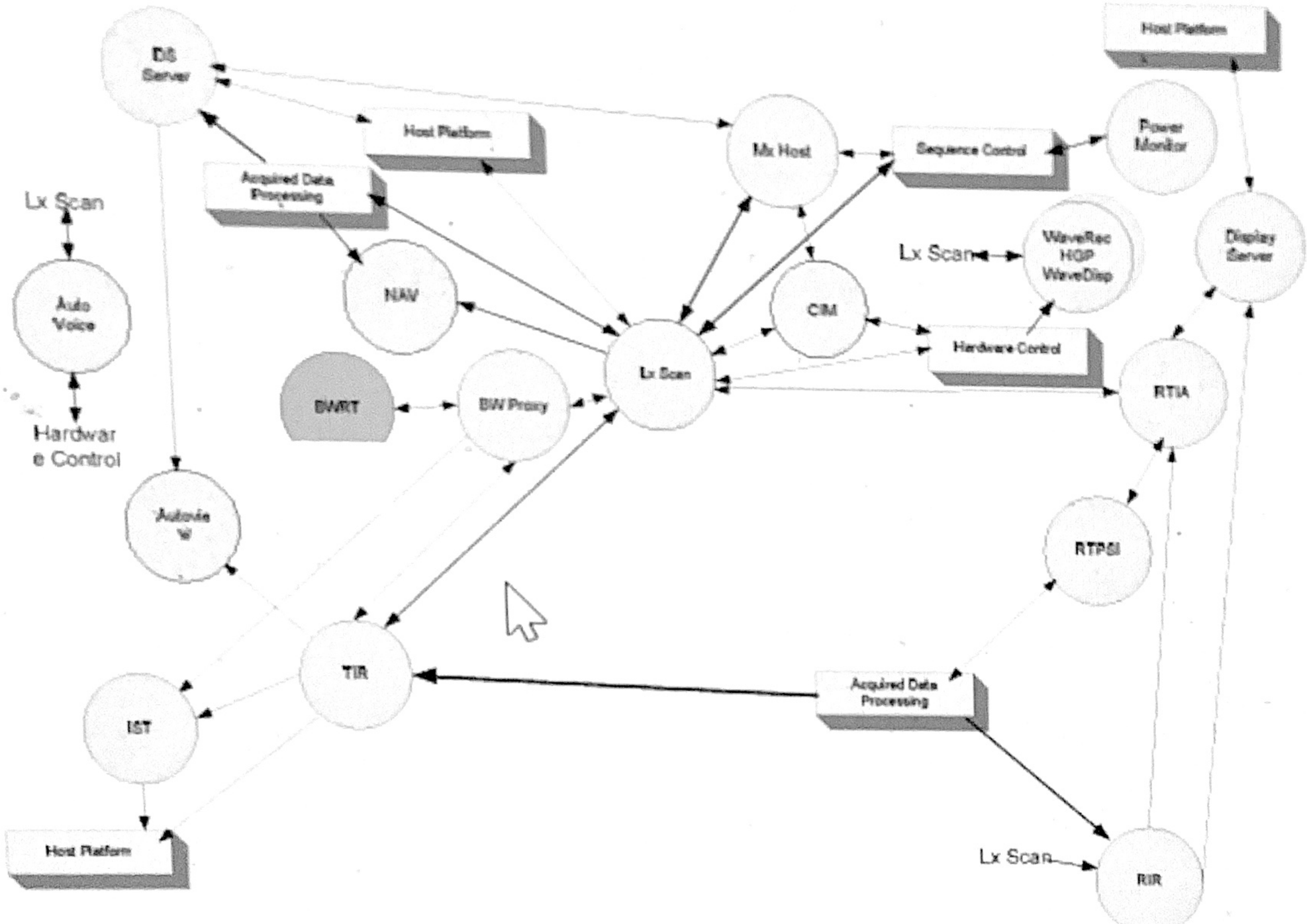
- Lx Scan loads appropriate target PSD into sequencer
- Lx Scan requests Host PSD to transfer all prescription information into target PSD
- CIM calculates hardware settings based on prescription and downloads them to hardware control
- Based on operator choice Host Prescan either requests acquired data processing to calculate hardware settings for maximum output or provides user interface for operator to set hardware values. It also requests auto coil algorithm
- "For dynamic coil configuration, host prescan runs full coil mode calibration if not available.
- For spectro applications spectro prescan provides user interface for operator to set hardware values.
- IFCC generates image headers based on user prescription values and communicates image reconstruction parameters to acquired data processing.
- Based on operator choice start/stop/pause of scanner can be initiated from user interface or hardware control. In both cases scanner state is communicated between Lx Scan and hardware control via acquired data processing.

For Mx tasks

- Lx Scan sends download request to Mx JVM
- Mx JVM downloads information to Hardware Control. Acquired data processing and DsServer
- Based on operator choice Mx JVM either requests acquired data processing to calculate hardware settings for maximum output or provides user interface for operator to set hardware values.
- Based on operator choice start/stop/pause of scanner can be initiated from user interface or hardware control. In both cases scanner state is communicated between Mx JVM and hardware control.

3.4.1.4 Scan Execution

Figure 11 Data Paths for Scan Execution



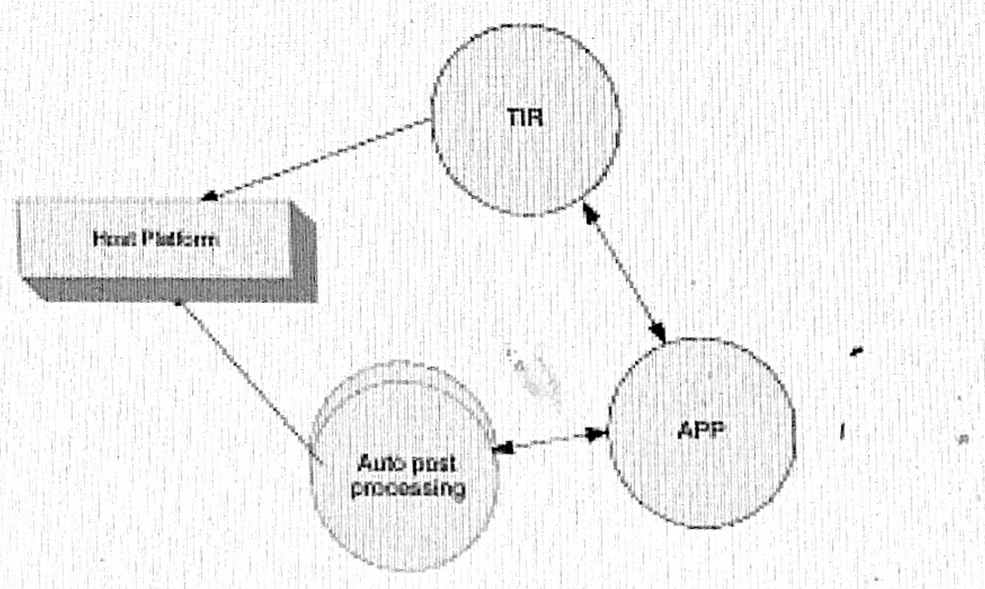
Scan execution is controlled within the hardware control, sequence control and acquisition processing subsystems as well as the host software subsystem. The following key actions occur to complete this processing step:

- The pulse sequence manages the sequence control hardware directing the play out of the Gradient and RF waveform and digitization of the k-space data.
- The data handling component of the reconstruction subsystem receives the k-space data
- The acquired data processing component of the reconstruction subsystem generates image pixel data from the collected k-space data
- The acquired data processing component of the reconstruction subsystem transfers the image pixel data to the host software subsystem

- For Lx applications TIR receives pixel data from acquired data processing unit and generates DICOM image by combining the saved image header parameters and pixel data
- For Mx applications DsServer receives pixel data from acquired data processing unit and generates DICOM image by combining the saved image header parameters and pixel data
- Except for FMRI and RTIA applications, TIR/DsServer transfers the DICOM images to the host platform for storage into the image database. They also write images into shared buffer to display in autoview.
- Host platform stores DICOM image into database for further processing or future use.
- For real-time applications, RTIA JVM displays the acquisition control user interface to allow operator interaction and/or acquisition monitoring. RTPSI Server communicates user operation to sequence control. RIR receives pixel data from acquired data processing and generates DICOM image. Display Server displays the real time images. Based on operator request, it also sends images into Host platform for inserting into database.
- For FMRI applications, TIR receives pixel data from data processing unit. It generates DICOM image and sends them to BWRT for further processing and display. BWRT-proxy acts as communication interface between host software and BWRT. After processing, BWRT send images to IST, which in turn sends images to host platform for inserting into database.
- For MR-Echo applications, Mx JVM displays the acquisition control user interface to allow operator interaction and/or acquisition monitoring. It also provides mechanism to switch into high resolution scans on fly. It is also possible to prescribe graphically for these scans using real time localizers. User has flexibility to bookmark the real time images and save them to database.

3.4.1.5 *Automated Post Processing*

Figure 12 Data Paths for Post processing

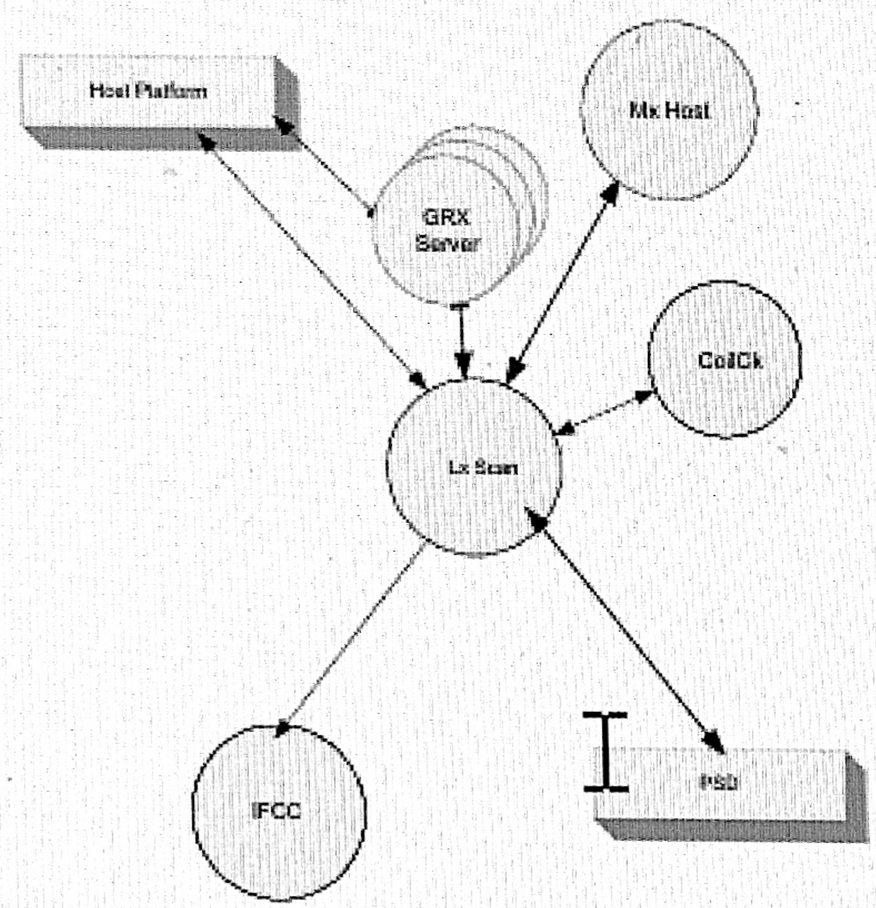


The In-line post processing is initiated by the host software subsystem. The following key actions occur to complete this processing step:

- As TIR receives images, it checks if it needs post processing. It sends notification to APP on arrival of image that needs post processing.
- APP queries images from database and performs post processing
- APP generates post processed images and send them to host platform for inserting into database.

3.4.1.6 *Exam Termination*

Figure 12 Data Paths for Exam termination



Exam termination is controlled by host platform subsystem and host software subsystem.

Host platform provides interface for terminating the exam. Following key actions occur in this step:

- Host platform communicates "End Exam" action to Lx Scan.
- Lx Scan closes graphical and prescription user interfaces
- Lx Scan terminates all host PSDs.
- Lx Scan releases memory for all the data associated with the exam.
- Based on user preferences Lx Scan starts automatic network & archive operations.

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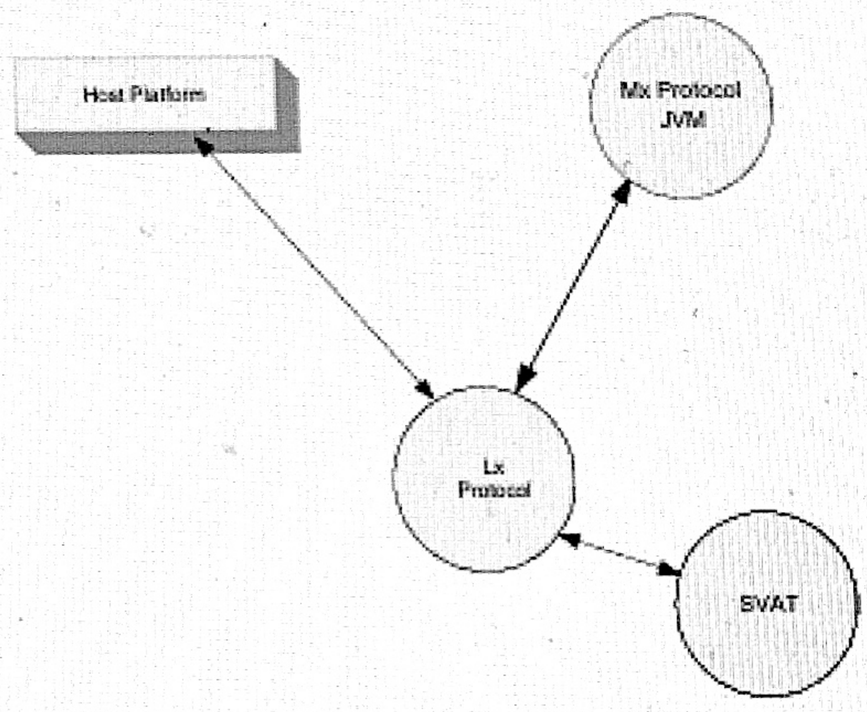
- IFCC/CIM cleans-up all data associated with the Exam.
- Lx Scan terminates Mx JVM associated with this exam. Host platform starts new Mx JVM.
- Lx Scan sends notification to acquired data processing to trigger the BAM clearing.

3.4.1.7 Protocol Session

Figure 12 Data Paths for Protocol Session



Protocol session is managed by both Host platform subsystem and Host software subsystem. In this session operator can create or modify site protocols.

- Host platform displays protocols and loads protocol selected by operator
- Lx Protocol provides prescription screen for Lx tasks to change parameters in protocol.
- Mx Protocol JVM provides prescription screen for Mx tasks to change parameters in protocol
- Lx Protocol provides interface to store protocols

3.4.2 Preprocessing

Preprocessing with respect to this subsystem occurs during system initialization, largely driven by the host platform. During host software initialization the following actions occur:

- It reads MR configuration file, Gradient configuration file, Host configuration file, Suite Configuration file, Pure configuration file and SAR/dbdt configuration file.
- It does initialization for different communication channels. Based on communication channel used by component, it does CUP initialization, mrmail initialization, CORBA initialization and

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GEOCOM initialization. They also subscribe to mrmail packets, corba events and GEOCOM services.

- It identifies the mode to start. Reads configurations files and environment variables to switch into appropriate mode.
- It creates global parameter and CV objects. Initializes all global variables. Motif based applications create UI components after reading properties files. Applications that control UI, creates table that map UI component to call back method.
- It removes old temporary files if they exist . It prunes the old log files and starts new log file.
- It checks the status of other applications and also checks hardware. It updates the user interface based on this status.
- On completing startup, each process will inform host platform.

3.4.3 Initialization

Initialization with respect to this subsystem occurs at the beginning of new exam startup. This area was previously covered in Section 3.4.1.1

3.4.4 Termination

Termination with respect to this subsystem occurs during end exam. This area was previously covered in Section 3.4.1.6

3.4.5 Error Handling and Recovery

There are three major classes of errors handled within the host software subsystem:

- Automatically Recoverable errors – such as auto restarting of a server (i.e. GRx server)
- User recoverable errors – errors that the user can recover via retrying the operation or requesting a subsystem reset (i.e. TPS reset)
- Unrecoverable errors – errors that cause an automatic system shutdown.

3.5 Data Design

The data managed by the host subsystem fall in to the following general categories: static data storage, and dynamic data storage.

3.5.1 Static Data

The primary static data associated with the host software subsystem is user pulse sequence data, auto transfer configurations, auto voice configurations, user interface labels, error messages and protocol data.

The pulse sequence data has details about each pulse sequence button describing its family, label, annotation and default imaging options. The auto transfer configuration has details about auto transfer and auto archive options along with host names. Auto voice configuration data consists of audio files in all supported languages. Label data consists of text displayed on all user interfaces in all supported languages. Error message data consists of all the error messages logged into system log file in all supported languages. The protocol data exists in three primary categories: GEHC, Service, and Site. The GEHC and Service categories are not user modifiable, the site category is user created and managed.

Host software subsystem reads magnet, table and other hardware details from MR configuration file. It reads gradient information from Gradient configuration file. For twin system, based on user selection of gradient mode appropriate gradient configuration file is read. It reads SAR limits from SAR configuration file. It reads dbdt limits from DBDT configuration file. It reads site specific information like hospital name, date format, time format, keyboard type, language, weight unit, temperature unit etc. from suite configuration files. Details like system ID and service ID are read from host configuration file. pure specific settings are read from PURE configuration file. User interface specific settings are read from scan configuration files. Recon performance levels and color codes are read from recon configuration file. Coil related information is read from coil database. It reads dual drive parameters from dual drive configuration file. Prescan related setting are read from prescan configuration file.

3.5.2 Dynamic Data

The primary dynamic data associated with the host software subsystem is the data associated with the individual series being prescribed in an exam, image header data, waveform data, prescan data and image data.

The data associated with series consists of prescribed parameter information, graphical prescription information, host computed information, and psd computed information. Image header information consists of patient information, prescription information, host computed information and psd computed information. Waveform data consists of information associated with waveform display, scan counter and service key. Prescan data consists of hardware settings for specific application or coil. Image data consists of DICOM image header along with pixel information.

4. Detailed Design

Some components described in the Design Architecture section (3) will require a more detailed design depending on how critical they are (it is mandatory for **class C** components). Each subsection of this section will refer to or contain a detailed description of a design output component requiring detailed design.

4.1 Lx Scan

The detailed design for the Lx Scan is covered in the Lx Shell SWDD, ref [6]

4.2 Mx JVM

The detailed design for the Lx Scan is covered in the DV25 Host Propeller SWDD, ref [17]

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4.3 *CIM*

The detailed design for the CIM is covered in Coil Data Management SWDD, ref [8]

4.4 *SVAT*

The detailed design for the SVAT is covered in Special Optional SVAT Support SWDD, ref [18]

4.5 *Prescan & SpectroPS*

The Prescan and Spectro Prescan components are legacy components and no further documentation on the components design is available at this time.

4.6 *GRX Server (2 sets) and Localizer Viewer*

The detailed design for the GRX Server component and localizer viewer is in Graphic Rx SWDD [7]

4.7 *GRx Proxy*

The GRx Proxy component is a legacy component and no further documentation on the components design is available at this time.

4.8 *Autoview*

The Autoview component is a legacy component and no further documentation on the components design is available at this time.

4.9 *DS Server*

The detailed design for DS Server component is in Image Receive Chain SWDD [9]

4.10 *BW Proxy*

The detailed design for the brainwave proxy is in Brain Wave SWDD [13]

4.11 *RXM*

For detailed design for RXM, refer section 5.4.2 in MNI technical reference file [22]

4.12 *BWRT*

For detailed design for BWRT, refer section 5.4.2 in MNI technical reference file[22]

4.13 *IFCC*

The detailed design for IFCC component is in Image Receive Chain SWDD [9]

4.14 *TIR*

The detailed design for TIR component is in Image Receive Chain SWDD [9]

4.15 *IST*

The detailed design for IST component is in Image Receive Chain SWDD [9]

4.16 *APP*

The detailed design for APP component is in Image Receive Chain SWDD [9]

4.17 *RTIA*

The RTIA component is a legacy component and no further documentation on the components design is available at this time.

4.18 *Display Server*

The Display Server component is a legacy component and no further documentation on the components design is available at this time.

4.19 *RIR*

The RIR component is a legacy component and no further documentation on the components design is available at this time.

4.20 *RTPSI*

The RTPSI component is a legacy component and no further documentation on the components design is available at this time.

4.21 *ILT*

The ILT component is a legacy component and no further documentation on the components design is available at this time

4.22 *NAV*

The detailed design for NAV component is in Host new Enhancements SWDD [15]

4.23 *Auto Voice*

The detailed design for Auto Voice component is in Auto Voice SWDD [19]

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4.24 *Power Monitor*

The detailed design for Power Monitor component is in Host new Enhancements SWDD [15]

4.25 *In Room Monitor*

The detailed design for In Room Monitor component is in In Room Monitor SWDD [11]

4.26 *Express Spine Plugin*

The detailed design for Express Spine Plugin component is in In Express Spine Annotation SWDD [16]

4.27 *One Touch Spine Library*

The detailed design for One Touch Spine Library component is in In Express Spine Annotation SWDD [16]

4.28 *EXSI*

The detailed design for the EXSI is covered in EXSI SWDD, ref [20]

4.29 *Dual Drive*

The detailed design for the Dual Drive is covered in Dual Drive SWDD, ref [20]

5. Software Generation

This section describes the top level look at the software generation relative to tools used across subsystems and major SOUP components. The software source files as well as the compilers and system libraries are maintained under clearcase revision control. The build workstations are capable of executing the cross-compilers to produce the object files for the hosting operating system on each of the specific subsystem computers. Refer to the design documents for the actual locations and building mechanisms used for each component.

5.1 *Tools Used*

Following tools are used by host software sub-system

5.1.1 UIMX UI Builder

This is used for building X-motif user interface.

5.1.2 TAO IDL Compiler

This is used for compiling CORBA idl files into C++.

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5.1.3 OPENORB IDL Compiler

This is used for compiling CORBA idl files into JAVA.

5.1.4 JAVA Compiler

This is used for compiling Java files.

5.2 Files Described

This section lists directory path to the location of source files for the associated components.

Lx Scan:

/vobs/scan/MrScan

/vobs/host/foundation-libs/StateFiles

/vobs/host/foundation-libs/Tim

/vobs/host/corba-libs

/vobs/host/foundation-libs/uid

/vobs/host/svat_objc

/vobs/host/WorkflowMgr/components/gemriwfm/components/lxscn

GRXServer:

/vobs/scan/3PlaneGRx

Geometry:

/vobs/scan/SrxGeosUtil

Mx JVM:

/vobs/mxui

/vobs/mxframework

/vobs/mxapp

/vobs/mxhardware

/vobs/mxutil

/vobs/mxtools

/vobs/mxproduct

CIM:

/vobs/scan/MrScan/CoilID

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SVAT:

/vobs/host/svat

LAIS:

/vobs/host/foundation-apps/Lais

ILT

/vobs/host/foundation-apps/ImageListener

Host Prescan:

/vobs/host/prescan

Spectro Prescan :

/vobs/host/foundation-apps/hostSpectro
/vobs/host/foundation-libs/hostSpectroCtrl
/vobs/host/foundation-libs/hostSpectroUIF
/vobs/host/foundation-libs/hostSpectroUtil

Navigator:

/vobs/scan/MrScan/Navigator

Brainwave:

/vobs/host/fmri

Power Monitor:

/vobs/scan/MrScan/ PowerMon

IFCC:

/vobs/host/foundation-apps/IFCC

/vobs/host/foundation-libs/IMS

/vobs/host/foundation-libs/uid

TIR:

/vobs/host/foundation-apps/TIR

/vobs/host/foundation-libs/lx2dicom

/vobs/host/foundation-libs/uid

/vobs/host/foundation-libs/IMS

DSServer:

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/vobs/mxproduct/mxservers/ds_host

/vobs/mxds/gemrimxds

IST:

/vobs/host/fmri

APP:

/vobs/host/AutoSubtract

/vobs/host/DTI

/vobs/host/foundation-apps/APP

Autoview:

/vobs/host/foundation-apps/Autoview

RTIA:

/vobs/RTIA/java

Display Server:

/vobs/RTIA/displayserver

/vobs/RTIA/gvtk

RIR:

/vobs/RTIA/rir

/vobs/host/foundation-libs/lx2dicom

RTPSI:

/vobs/RTIA/rtpsi

HGP:

/vobs/scan/hgp

WaveDisp & WaveRec:

/vobs/host/WaveDisplay

AutoVoice:

/vobs/host/autovoice

Dual Drive:

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/vobs/host/WorkflowMgr/components/gemriwfm/components/services/DualDriveService.java

/vobs/scan/MrScan/CoilID/cim/DualDriveService.cc

/vobs/host/presel

General

/vobs/scan/include

/vobs/host/include

/vobs/lx/include

5.3 SOUP Used

SOUP (Software of Unknown Pedigree) is needed to support the proper functioning of the host software defined by this design document. Following is a list of the SOUP items and where their functional, performance, and system software and hardware resource requirements are specified.

5.3.1 JAVA

Java programming language (a high-level, object-oriented programming language); the Java Virtual Machine (a high-performance virtual machine that executes byte codes on a specific computing platform, typically abbreviated JVM); and the Java platform, a JVM running compiled Java byte codes, usually calling on a set of standard libraries such as those provided by Java Standard Edition (SE). Please refer [24] for this SOUP item.

5.3.2 ACE/TAO, Open ORB

Adaptive Communication Environment (ACE) is a freely available, open-source, object-oriented (OO) framework that implements many core patterns for concurrent communication software. ACE provides a rich set of reusable C++ wrapper facades and framework components that perform common communication software tasks across a range of OS platforms. The communication software tasks provided by ACE include event de-multiplexing and event handler dispatching, signal handling, service initialization, inter process communication, shared memory management, message routing, dynamic (re)configuration of distributed services, concurrent execution and synchronization.

TAO is an open-source implementation of a CORBA Object Request Broker (ORB) built using components and patterns in the ACE framework. Open ORB provides a Java implementation of the OMG CORBA 2.4.2 specification. The ACE, TAO and Open ORB components are associated with the communication between the host components associated with prescription and data acquisition. Please refer [23] for this SOUP item.

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-%20GE%20HealthCare/Documents/02MR_Host/Host_Software_SSD_DOC265999

69 of 73

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5.3.3 JGOODIES

The JGoodies suite provides components and solutions that complement Swing to solve common user interface tasks. It advocates a UI production process that lets you save time and money while ensuring consistent and elegant design. Please refer [27] for this SOUP item.

5.3.4 JGo

JGo is a Java user interface component that make it easy to include interactive diagrams as part of your Java application or applet user interface. JGo provides creating graph / waveform for manual prescan. Please refer [26] for this SOUP item.

5.3.5 XERCES

XERCES is a library for parsing, validating and manipulating XML documents. These components are associated with the Mx application user interface. Please refer [28] for this SOUP item.

5.3.6 XRT

XRT/Graph is a powerful widget that can display two dimensional graph that are used by Lx Manual Prescan, Navigator and Spectro Prescan in Host software. Please refer [29] for this SOUP item.

5.3.7 Merge DICOM ToolKit

Merge DICOM toolkit is used to read or write DICOM objects. TIR and RIR components use this in creating DICOM image. Please refer [30] for this SOUP item.

6. Traceability

This design document uses DOORS for traceability. Each requirement section that this design document covers is traced via a DOORS object, which corresponds to the document. See program design traceability DHF requirements for more information about.

Also see section 5.2 for directories/files traceable to this design.

7. Glossary

Term	Definition
ADS	Acquisition data server
CORBA	Common Object Request Broker

Term	Definition
ADS	Acquisition data server
CSV	Computerized System Validation
DAB	Data Acquisition Board
GOC	Global Operator Console
i18n	Internationalization
ICN	Image Compute Node
MPPS	Modality Performed Procedure Step
PACS	Picture archive system
ROI	Region of Interest
RTP	Real-time processing
SAT Band	Saturation Band
SOUP	Software of Unknown Pedigree
SVAT	System Validation and Test
PSD	Pulse Sequence
DICOM	The Digital Imaging and Communications in Medicine
TCP/IP	Transmission Control Protocol/Internet Protocol
POSIX	Portable Operating System Interface
CUP	Common Unix Platform
MX JVM	Monarch Java Virtual Machine
IFCC	Image File Creation and Cleanup
TIR	Tardis Image Receive
APP	Auto Post Processing
RTIA	Real Time Image Acquisition
RIR	Real time Image Receive
RTPSI	Real Time TPS Interface
NAV	Navigator
ILT	Image Listener
FMRI	Functional Magnetic Resonance Imaging

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Term	Definition
ADS	Acquisition data server
HGP	Host Gating Process
IPC	Inter Process Communication
UDP	User Datagram Protocol
VRE	Volume Reconstruction Engine
ACE	Adaptive Communication Environment
JAI	Java Advanced Imaging
ORB	Object Request Broker
GEOCOMM	GE Object Communication
EXSI	External Scanner Interface
PXE	Paradigm eXecution Engine